



Inspiring Excellence

Assignment-04

Sequential Circuit

Name: Md. Tahsin Hossain

ID: 24301099

Course: CSE260

Section: 17B

Submitted to: Majisha Jahan Disha [MAJJ]

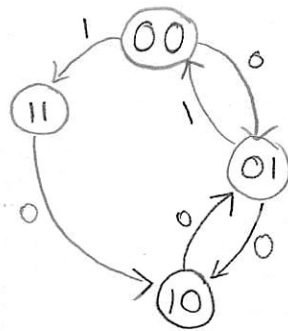
Ans. to the ques. no-01

Action movies $\rightarrow$ AM (00)

Romantic movies $\rightarrow$ RM (01)

Comedy movies $\rightarrow$ CM (10)

Documentary movies $\rightarrow$ DM (11)



Present State			Next state		Flip-flop inputs			
A	B	n	A ⁺	B ⁺	J _A	K _A	J _B	K _B
0	0	0	0	1	0	X	1	X
0	0	1	1	1	1	X	1	X
0	1	0	1	0	1	X	X	1
0	1	1	0	0	0	X	X	1
1	0	0	0	1	X	1	1	X
1	0	1	X	X	X	X	X	X
1	1	0	1	0	X	0	X	1
1	1	1	X	X	X	X	X	X

A \ Bx	00	01	11	10
0		1		1
1	X	X	X	X

$$J_A = \bar{B}x + B\bar{x}$$

$$= B \oplus x$$

A \ Bx	00	01	11	10
0	X	X	X	X
1	1	X	X	

$$K_A = \bar{B}$$

A \ Bx	00	01	11	10
0	X	X	1	1
1	X	X	X	1

$$K_B = 1$$

A \ Bx	00	01	11	10
0	1	1	X	X
1	1	X	X	X

$$J_B = 1$$

